Curriculum Vitae

MD MINHAZUL HAQ

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Professional Experience

Graduate Research Assistant (GRA) *Nov 2023 – Jun 2025*

Centre of Intelligent Systems for Emerging Technology (CISET) Lab, Department of Electrical Engineering, Faculty of Engineering, Universiti Malaya (UM)

- Developed and optimized lightweight YOLO-based object detection models for accurate and efficient multi-scale object detection and classification on resource-constrained devices.
- Designed efficient architectures for real-time object detection on embedded platforms (e.g., NVIDIA Jetson Nano), improving inference speed and detection accuracy for practical deployment in smart surveillance and IoT applications.

Part-time Teaching, Lab Demonstrator / Instructor *Oct 2025 – Feb 2026*

LAB 2 (KIE1002), Department of Electrical Engineering, Faculty of Engineering, Universiti Malaya (UM)

- Conducted lab sessions for undergraduate engineering students, demonstrating and explaining key electrical and electromagnetic experiments (circuit laws, amplifiers, diode clipping, magnetic induction, Lorentz force, and PCB layout design) while maintaining standard safety protocols.
- Guided students to conduct successful experiments and analyse results (voltage, current, capacitance, inductance, power, gain, etc.) and evaluated their practical understanding through interactive discussions and interview session.
- Troubleshooting experimental devices including Function Generators (Analog and Digital), Digital Oscilloscope, Multimeters during the lab sessions.

Project Work as a Graduate Research Assistant (GRA)

- "Sustainable And Smart Food Packaging System, IMG001-2022", Supervised by Associate Prof. IR. Dr. Anis Salwa Binti Mohd Khairuddin and Dr. Effariza Hanafi, Universiti Malaya (UM), Nov 2023 - Jun 2025.
- "Algae Cultivation" and "Fabrication of an Electrochemical Sensor with Nanocomposite Sensing Layer for Hydrogen Peroxide Detection" Supervised by IR. Dr. Suriani Binti Ibrahim and Associate Prof. IR. Dr. Anis Salwa Binti Mohd Khairuddin, Universiti Malaya (UM), Nov 2024 – May 2025.
- "Arfes: Augmented Reality-based Educational Framework for Engineering Students, UMG002-2025ASPD", Supervised by Dr. Effariza Hanafi, Universiti Malaya (UM), Nov 2025 – Feb 2026.

Standardized Test Scores

Test / Date	Overall	Speaking	Listening	Reading	Writing
IELTS – 8 Apr 2025	7.0	7.5	7.5	7.0	6.0
IELTS – 23 Jun 2023	6.0	6.5	6.5	5.5	6.0

Publication

Postgraduate

1. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi, Hafiz Muhammad Fahad Noman, Mohamad Haniff Junos. HAT-YOLO: Hybrid Attention and Transformer-Integrated Model for UAV Object Detection on Embedded Devices. (KSII TRANSACTIONS ON INTERNET AND INFORMATION SYSTEMS VOL. 20, NO. 2, Feb. 2026).
2. Yang Z, Khairuddin AS, Huang CJ, Ru WW, Noman HM, Putri TW, Xu X, Haq MM. A Review of Short-Term, Long-Term, and Memory Consolidation Mechanisms in the Hippocampus and Cerebral Cortex. IEEE Access. 2025 Mar 27.
3. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. EdgeLite-YOLOv8n: An Efficient YOLOv8n Model for Real-Time Vehicle Detection in UAV Imagery on Embedded Platforms. Submitted to 2nd International Transdisciplinary Research Conference 2026, Universiti Malaya.
4. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. Improved YOLOv8n model for Automated Wood Texture Defect Detection based on Computer Vision (Ready to submit).

Undergraduate

1. Hasan, M. M., K, S., A, M., & Haq, M. M. (2024). DEVELOPMENT OF PROTOTYPE DUAL-AXIS SOLAR TRACKING DEVICE FOR OPTIMAL POWER GENERATION AT RESIDENTIAL AREA IN BANGLADESH. Journal of Energy and Safety Technology (JEST), 7(1), 25-33

Academic Credentials

Master of Engineering Science (Research) *Nov 2023 – Ongoing*
Department of Electrical Engineering, Faculty of Engineering, Universiti Malaya (UM)
Field of Study – Artificial Intelligence
Thesis title – *Optimization of Lightweight YOLO Model for Automated Object Detection on Embedded Device*

Bachelor of Electrical and Electronic Engineering *1st Jan 2017- 6th Apr 2022*
Jashore University of Science and Technology (JUST) *GPA: 3.11/4.00*
Major: Electrical Engineering, Minor: Machine Learning
Thesis tittle – *Design of Solar Tracking System for Optimal Power Generation at Residential Area in Bangladesh*

A level – Nawabganj Govt. College, Bangladesh *GPA: 5.00/5.00 (2016)*
O level – Harinagar High School, Bangladesh *GPA: 5.00/5.00 (2014)*

Awards and achievements

- ***Best Presenter (Postgraduate Oral Presentation)-Winner*** – 4th International Orthopaedic Postgraduate Symposium (IOPS4), Faculty of Medicine, Universiti Malaya, 29 Oct 2025. (Title: "*Lightweight YOLOv8 with Attention-Enhanced Backbone for Real-Time Paediatric Wrist Fracture Detection Using Resource-Constrained Devices*").
- ***Best Project Award – Science Fair 2019***, Jashore University of Science and Technology (JUST), Bangladesh, Mar 2019. (Project: "*Humanoid Robot*" – a microcontroller-based, voice-controlled robot capable of multi-action movements.)
- ***Best Project Award – Student to Startup 2.0 Campus Hunt,*** Jashore University of Science and Technology (JUST), Bangladesh, Oct 2019. (Project: "*Solar Scooter*" – 150 kg load capacity with dual AC/Solar charging capability.)

Poster presentation

- Md. Omar Faruk, Md. Minhazul Haq, Anis Salwa B. M. Khairuddin, Effariza Hanafi, Karen L. Pua, and W. M. Tilakaratne. "Real-Time Oral Cancer Detection Using Lightweight YOLOv8n-Based Deep Learning and Computer Vision: A Pilot Study." UMHEALTH - Academic Month & Research Carnival 2025, Universiti Malaya.

Extra-curricular activities

- Organizing Secretary, JUST Robo Society, Jashore University of Science and Technology (JUST), Mar 2020 – Mar 2021.
- Organizer, Seminar on Artificial Intelligence and Machine Learning, Jashore University of Science and Technology (JUST) 2020.
- Organizing Committee Member, HULT Prize at JUST, Jashore University of Science and Technology (JUST), 2018.

Technical Skills

Programming Languages:	Python, C, C++
Computer vision model:	YOLO (v26, v12, v11, v10, v9, v8, v7, v5), CNN, RNN, SSD
Engineering & Analytical tools:	MATLAB, PSpice, Proteus, CST, Power BI
Office tools:	MS Word, MS PowerPoint, MS Excel

Linguistic Proficiency

English (Excellent Proficiency)
Bangla (Native Language)
Bahasa Melayu (Basic)

References

Associate Prof. Ir. Dr. Anis Salwa Binti Mohd Khairuddin (HOD)
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Faculty of Engineering, Universiti Malaya UM)
Contact No : 0379675205
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Dr. Effariza Hanafi
Dept. of Electrical Engineering
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